Lab Report: Model comparison on benchmarks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 12 GB of memory.

2 Results

Figure 1 shows the f1 score of ResNet-18 by dataset.

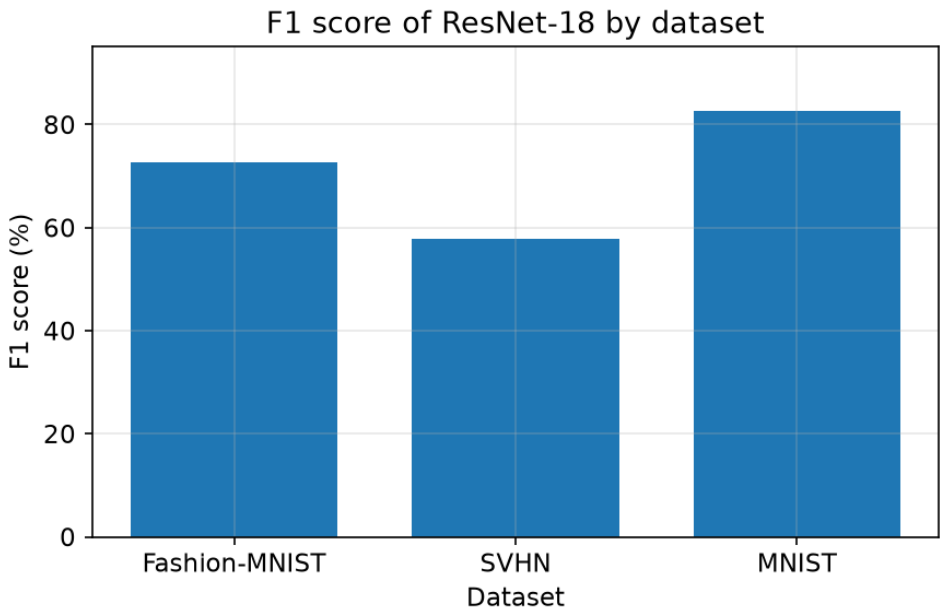


Figure 1: F1 score of ResNet-18 by dataset for the models in this experiment.

As shown in Figure 1, ResNet-18 reaches 80.8% F1 score on MNIST. ResNet-18 scores 10.0 points higher on MNIST than on Fashion-MNIST (Figure 1). As shown in Figure 1, on SVHN, ResNet-18 obtains a F1 score of 76.9%.

Figure 2 shows the validation accuracy by dataset.

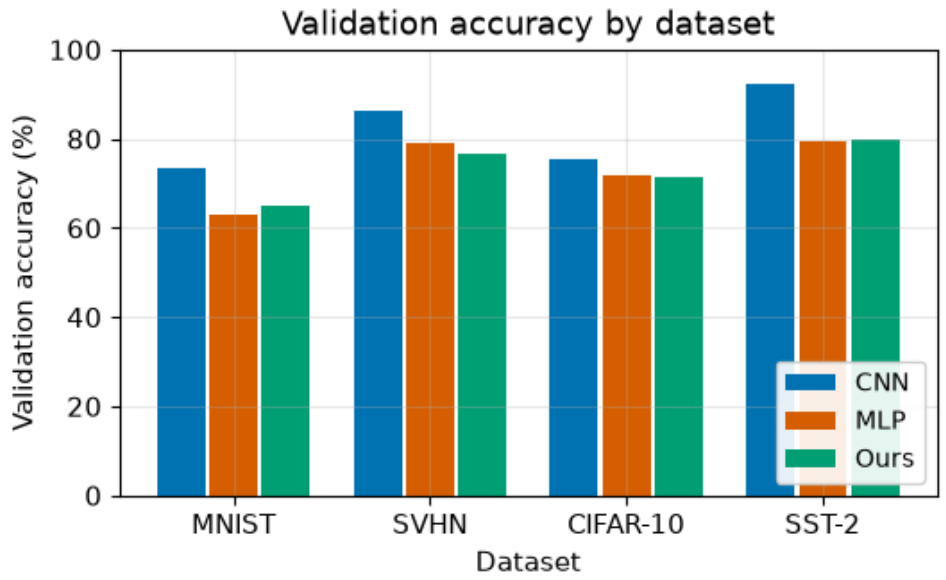


Figure 2: Validation accuracy by dataset for the models in this experiment.

MLP achieves 75.3% validation accuracy on CIFAR-10. As shown in Figure 2, on CIFAR-10, CNN is 13.9 points better than Ours. Ours reaches 65.1% validation accuracy on MNIST (Figure 2).

Figure 3 shows the f1 score by dataset.

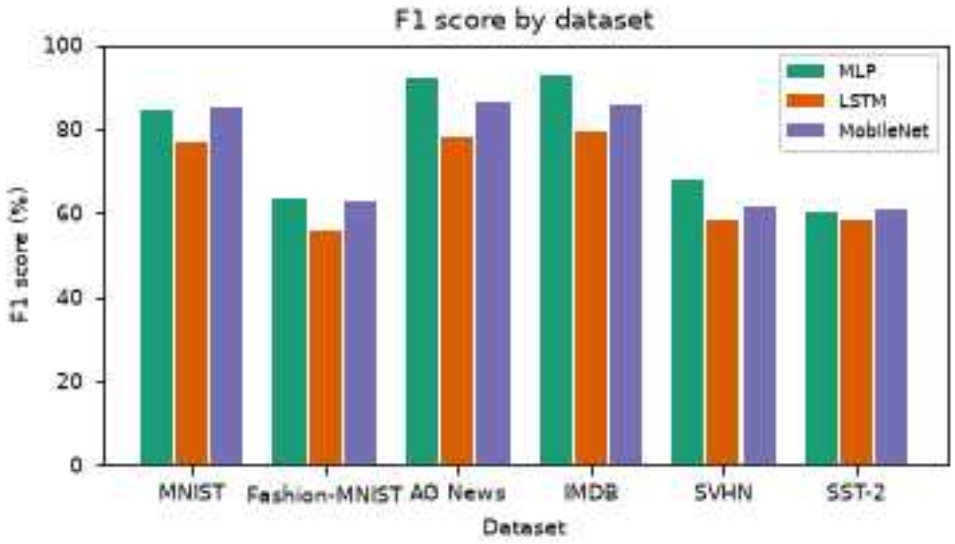


Figure 3: F1 score by dataset for the models in this experiment.

On SVHN, MLP obtains a F1 score of 68%. On MNIST, MLP obtains a F1 score of 77.1%. As shown in Figure 3, MobileNet achieves 60.6% F1 score on SST-2. MLP reaches over 55% F1 score on SST-2.

3 Discussion

We note that the training was stable for all models.